Maths —— Assignment-3

1. $A = \begin{bmatrix} 4 & 2 \\ 1 & 3 \end{bmatrix}$ $\lambda I = \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}$

$$A - \lambda I = \begin{bmatrix} 4-\lambda & 2 \\ 1 & 3-\lambda \end{bmatrix}$$

$\Rightarrow |A - \lambda I| = 0$

$(4-\lambda)(3-\lambda) - 2 = 0$

$12 - 4\lambda - 3\lambda + \lambda^2 - 2 = 0$

$\lambda^2 - 7\lambda + 10 = 0$

$(\lambda - 5)(\lambda - 2) = 0 \Rightarrow \lambda = 5, 2$.

for $\lambda = 5$

$A - \lambda I = \begin{bmatrix} 4-5 & 2 \\ 1 & 3-5 \end{bmatrix}$

$= \begin{bmatrix} -1 & 2 \\ 1 & -2 \end{bmatrix}$

$\Rightarrow -x + 2y = 0$

$\Rightarrow x = 2y$

Eigen vector $= \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

for $\lambda = 2$

$A - \lambda I = \begin{bmatrix} 4-2 & 2 \\ 1 & 3-2 \end{bmatrix}$

$= \begin{bmatrix} 2 & 2 \\ 1 & 1 \end{bmatrix}$

$\Rightarrow x + y = 0$

$\Rightarrow x = -y$

Eigen vector $= \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

Eigen values $(\lambda) = 5$ & 2

Eigen vectors $= \begin{bmatrix} 1 \\ 2 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

Q2. $A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 4 \\ 0 & 4 & 9 \end{bmatrix}$ $\lambda . I = \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix}$

$$A - \lambda I = \begin{bmatrix} 2-\lambda & 0 & 0 \\ 0 & 3-\lambda & 4 \\ 0 & 4 & 9-\lambda \end{bmatrix}$$

principal equation

$$P(\lambda) = (2-\lambda)\left[(3-\lambda)(9-\lambda) - 4.4\right]$$
$$= (2-\lambda)\left[27 - 3\lambda - 9\lambda + \lambda^2 - 16\right]$$
$$= (2-\lambda)(\lambda^2 - 12\lambda + 11)$$

$\Rightarrow P(\lambda) = 0$

$(2-\lambda)(\lambda^2 - 12\lambda + 11) = 0$

$(2-\lambda)(\lambda - 11)(\lambda - 1) = 0$

$\boxed{\lambda = 2, 11, 1}$ Eigen values

for $\lambda = 2$

$$A - \lambda I = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 4 \\ 0 & 4 & 7 \end{bmatrix}$$

$0x = 0$

so x can be anything

$y = 0 \quad z = 0$

$$v_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

for $\lambda = 1$

$$A - \lambda I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 4 \\ 0 & 4 & 8 \end{bmatrix}$$

$x = 0 \quad y + 2z = 0$

$\Rightarrow y = -2z$

$$v_2 = \begin{bmatrix} 0 \\ -2 \\ 1 \end{bmatrix}$$

for $\lambda = 11$ $\qquad A - \lambda I = \begin{bmatrix} -9 & 0 & 0 \\ 0 & -8 & 4 \\ 0 & 4 & -2 \end{bmatrix}$

$\Rightarrow x = 0$

$\Rightarrow -8y + 4z = 0$

$y = 1 \quad z = 2 \qquad v_3 = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}$

8. SVD of $C = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix} \qquad C - \lambda I = \begin{bmatrix} 3-\lambda & 1 \\ 1 & 3-\lambda \end{bmatrix}$

$|C - \lambda I| = (3-\lambda)^2 - 1 = 0$

$\Rightarrow \lambda^2 - 6\lambda + 9 - 1 = 0$

$\Rightarrow \lambda^2 - 6\lambda + 8 = 0$

$\Rightarrow (\lambda - 4)(\lambda - 2) = 0 \Rightarrow \boxed{\lambda = 4, 2}$

$\lambda = 4$

$C - \lambda I = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}$

$x = y$

$\Rightarrow v = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$\lambda = 2$

$C - \lambda I = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$

$x = -y$

$v = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

$\Sigma = \begin{bmatrix} 4 & 0 \\ 0 & 2 \end{bmatrix} \qquad U = V = \frac{1}{\sqrt{1+1}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$

$C = U \Sigma V^T$

$C = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 4 & 0 \\ 0 & 2 \end{bmatrix} \left(\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \right)^T$

4. $D = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ $\quad D - \lambda I = \begin{bmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{bmatrix}$

$$|D - \lambda I| = (2-\lambda)^2 - 1 = 0$$

$$\Rightarrow \lambda^2 - 4\lambda + 4 - 1 = 0$$

$$\Rightarrow \lambda^2 - 4\lambda + 3 = 0$$

$$\Rightarrow (\lambda - 3)(\lambda - 1) = 0$$

$$\boxed{\lambda = 3, 1}$$

for $\lambda = 3$

$D - \lambda I = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}$

$x = y \Rightarrow v = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

for $\lambda = 1$

$D - \lambda I = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$

$x = -y \Rightarrow v = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

$P = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$ $\quad \Lambda = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}$ $\quad P^{-1} = \frac{1}{2}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$

$$D = P \Lambda P^{-1}$$

$$D = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix} \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

5. $f(x,y,z) = x^2y + yz^3$ at $(1,2,1)$

Gradient of $f = \nabla . f$

$$= \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z}\right)$$

$$= \left(2xy, x^2+z^3, 3yz^2\right)_{(1,2,1)}$$

$$\nabla f(1,2,1) = (4, 2, 6)$$

6. $\vec{F} = (x^2y)\hat{i} + (yz)\hat{j} + (xz^2)\hat{k}$

Divergence of $\vec{F} = \vec{\nabla} . \vec{F}$

$$= \frac{\partial}{\partial x}(x^2y) + \frac{\partial}{\partial y}(yz) + \frac{\partial}{\partial z}(xz^2)$$

$$\vec{\nabla} . \vec{F} = (2xy + z + 2xz)$$

7. $\vec{F} = (y^3)\hat{i} + (xz)\hat{j} + (xy)\hat{k}$

$$\text{curl}(\vec{F}) = \vec{\nabla} \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial_x & \partial_y & \partial_z \\ y^3 & xz & xy \end{vmatrix}$$

$$= \left(\frac{\partial xy}{\partial y} - \frac{\partial xz}{\partial z}\right)\hat{i}, \left(\frac{\partial xy}{\partial x} - \frac{\partial y^3}{\partial z}\right)\hat{j}, \left(\frac{\partial xz}{\partial x} - \frac{\partial y^3}{\partial y}\right)\hat{k}$$

$$\vec{\nabla} \times \vec{F} = (x - x)\hat{i}, (y - 0)\hat{j}, (z - 3y^2)\hat{k}$$

8. $f(x, y, z) = x^2 + y^2 + z^2$.

(i) Gradient of $f(x, y, z)$

$$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right)$$

$$= \frac{\partial\, x^2+y^2+z^2}{\partial x}, \frac{\partial\, x^2+y^2+z^2}{\partial y}, \frac{\partial\, x^2+y^2+z^2}{\partial z}$$

$$= (2x, 2y, 2z)$$

(ii) Divergence of gradient (Laplacian) $= \nabla \cdot (\nabla f)$

$$= \frac{\partial (2x)}{\partial x} + \frac{\partial (2y)}{\partial y} + \frac{\partial (2z)}{\partial z}$$

$$\nabla^2 f = 2 + 2 + 2 = 6$$